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Behavior in asymmetric dominance pairings predicts physiology, neural response, and neurogenesis in a social cichlid fish

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Introduction: In a social dominance hierarchy, dominant and subordinate individuals can exhibit dramatic differences in behavioral, physiological, and neural phenotypes. Juveniles often also form social dominance hierarchies, but it is less explored how these hierarchies reflect, or are distinct from, those exhibited by adults.

Methods: In the present study, we housed juvenile social cichlid *Astatotilapia burtoni* for one week in an asymmetric size pairing. We quantified behavior (approaches and displacements), stress (waterborne cortisol), neural activity (pS6), and neurogenesis (doublecortin) when the pair was reunited after a brief separation.

Results: Larger and smaller individuals differed dramatically in dominance behavior when reunited. Receiving these dominance behaviors did cause an acute stress response, but the asymmetric pairing itself did not induce chronic stress. We then assessed the impacts of this dominance pairing on the brain, and found that individuals who displayed more dominance behavior had lower neural activity when assessed across three forebrain regions. Lastly, a single week of pairing was not sufficient to induce changes in neurogenesis, but individuals who displayed more dominance behavior showed lower levels of neurogenesis.

Discussion: Like adults, juvenile *Astatotilapia burtoni* form dominance relationships after only one week of cohabitation. As the first to explore neurogenesis in juveniles of this species, we identified sites of particularly active neurogenesis and one region (the extended amygdala) as a potential novel site of juvenile-specific neurogenesis.

KEYWORDS

Astatotilapia burtoni, cortisol, development, doublecortin, neurogenesis, pS6, social dominance

Introduction

In many social species, individuals form consistent agonistic relationships known as dominance hierarchies. While the hierarchy structures vary across species and contexts, they reflect the general principle that individuals in a social group consistently vary in aggressive behaviors performed and received. These “pecking orders” are typically associated with

fitness-related outcomes (Strauss et al., 2022), and much of our understanding of the evolution of social dominance hierarchies comes from the reproductive benefits, costs, and skew in observed in adulthood (Ellis, 1995; Jandt et al., 2014). Dominant and subordinate individuals in a hierarchy can demonstrate drastic differences in behavior, physiology, and neural architecture, as detailed in Milewski et al. (2022). Dramatic changes can occur in circulating androgens (Beehner et al., 2009) and metabolism (Gilmour et al., 2017) as individuals establish and maintain hierarchy ranks.

These consistent differences in phenotype between dominant and subordinate individuals in a hierarchy often reflect prior social experiences. The early life environment contains a tapestry of interactions with parents (Champagne and Curley, 2005), siblings (Branchi et al., 2013), and unrelated members of larger social groups (White et al., 2002; Smith et al., 2010; Wallace et al., 2025) that shape adult behavior and physiology. For example, in Java sparrows (*Lonchura oryzivora*), juvenile birds “practice” courtship dances with other juveniles (Soma et al., 2019). While less explored than adult hierarchies, juveniles across many species also form dominance relationships prior to reproductive maturity. Crayfish, *Procambarus clarkii* (Girard) (Sato and Nagayama, 2012), bluebanded gobies, *Lythrypnus dalli* (Solomon-Lane et al., 2016), blue-footed boobies, *Sula nebouxii* (Drummond and Canales, 1998), yellow baboons, *Papio c. cynocephalus* (Pereira, 1989), and even humans (Thomsen, 2020) exhibit juvenile dominance hierarchies. Juvenile *Astatotilapia burtoni* (Burton’s Mouthbrooder cichlid) form social hierarchies prior to adulthood in as little as 30 minutes of interaction (Solomon-Lane and Hofmann, 2019; Solomon-Lane et al., 2022). In some cases, these early-life agonistic interactions directly predict phenotypes subsequently displayed as adults. For example, the social network position of juvenile male long-tailed manakins predict future reproductive success (McDonald, 2007). Conversely, even if the same behaviors can be observed across developmental stages, they may reflect distinct temporary niches with unique underlying selective pressures (Holekamp and Smale, 1998). One way to distinguish these two hypotheses is to ask: are the agonistic behaviors exhibited by juveniles and adults reliant on different underlying mechanisms?

A. burtoni are an excellent model to study the mechanisms of social dominance interactions, as they demonstrate a wide variety of changes to behavior, physiology, and neural architecture when transitioning flexibly between a subordinate to dominant phenotype (Fernald and Hirata, 1977; Maruska, 2014; Maruska and Fernald, 2018). Dominant and subordinate individuals differ in aggressive behavior, space use, testosterone (Friesen et al., 2022), and even cognitive performance (Wallace et al., 2022). In the brain, dominant and subordinate individuals differ in gonadotropin-releasing hormone neuronal properties (Greenwood and Fernald, 2004), in neural response in the basolateral amygdala, hippocampus, and in the lateral septum when engaging in a social encounter (Friesen et al., 2022). Given these comprehensive impacts of social dominance across biological systems in this species, in the present study, we assessed whether juvenile asymmetric dominance pairings would elicit similar effects to those observed in adults. Specifically,

we designed an experiment to test if short-term cohabitation in a dominance pairing as juveniles was sufficient to induce changes in aggressive behavior, physiology, neural response, and neurogenesis.

To do so, juvenile cichlids who had not yet shown visible signs of reproductive maturity were paired with another individual differing in size for one week. Conveniently, size is consistently used to induce asymmetric dominance pairings in this species (Alward et al., 2021; Solomon-Lane et al., 2022). After this week of cohousing, we conducted a separation-reunion assay where the two individuals were isolated from each other for a few hours and then reintroduced in order to observe agonistic behavior. Given that prior literature in this species found that early life rearing in different group sizes and group instability (via the introduction and removal of members) predicted agonistic behavior (Solomon-Lane and Hofmann, 2019; Harmon et al., 2024), we hypothesized that agonistic behaviors performed during the reunion phase would reflect the asymmetry in the dominance pairing as the two individuals re-established the dyadic hierarchy.

We also investigated whether asymmetric dominance pairings during the juvenile period would influence stress responses. In addition to the direct energetic costs of territorial aggression, fighting also elevates glucocorticoid levels during and after competition ends (Øverli et al., 2000; Ramos et al., 2021). While an acute stress response is generally considered an adaptive response to an immediate stressor, a repeatedly or continually activated stress response from competitive losses can have detrimental effects (Sapolsky, 1992; Gómez-Laplaza and Morgan, 2003; Sørensen et al., 2007). In adult *A. burtoni* specifically, dominant males show higher corticotropin-releasing factor mRNA than subordinates yet interestingly this does not correlate with circulating cortisol levels (Chen and Fernald, 2008). Supporting the conclusion that cortisol may not be the simple driver of agonistic behavior in this species, in a review on the correlation between dominance rank and cortisol levels in cichlid adults, Maruska et al. (2022) found no consistent relationship. Similarly, while early-life social experiences influenced a suite of behaviors that included open field exploration in juvenile *A. burtoni*, it did not significantly predict waterborne cortisol levels (Harmon et al., 2024). Taken together, these prior works suggests agonistic interactions may not induce effects of chronic stress, which we assessed in the current experiment.

Here we explored if asymmetric dominance pairings during the juvenile period influences the neural mechanisms underlying social decision-making. Two overlapping but distinct conserved neural circuits mediate social decisions, and they are highly conserved across species. The Social Behavior Network is a set of brain regions that exhibit bidirectional projections and mediate a variety of social behaviors, including aggression, parental care, and reproductive behavior (Newman, 1999; Goodson, 2005). Similarly, the Mesolimbic Reward System is a set of brain regions that are part of a conserved dopaminergic pathway and process stimulus salience (O’Connell and Hofmann, 2011; Adinoff 2004). These two networks function together in a collection known as the Social Decision-Making Network (SDMN) and has been characterized in mammals, birds, reptiles, amphibians, and teleosts (as extensively reviewed in O’Connell and Hofmann, 2011). We hypothesized that the neural mechanisms of agonistic behavior in juveniles would reflect those

employed in adults, and thus could be measured as differential neural activation between the two juveniles in an asymmetric size pairing. Here, we quantified the activation of the neural systems involved in social decision-making during the separation-reunion assay. Following the end of the assay, we extracted and processed brain tissue and immunohistochemically (IHC) labeled telencephalic (teleost forebrain) tissue for phosphorylated ribosomal protein S6 (pS6). pS6 is a marker that corresponds to an increase in recent translation (Magnuson et al., 2012; Butler et al., 2018) and pS6 IHC has previously been labeled to identify which cells have been recently activated in the cichlid brain (Butler et al., 2018; Jackson et al., 2025). In adult *A. burtoni* prior research has observed differential activation in multiple regions of the SDMN (basolateral amygdala, hippocampus, and lateral septum) during territorial encounters (Friesen et al., 2022). Here we expanded on this work to assess neural activation during social interactions in juvenile *A. burtoni*.

If the neural mechanisms of social decision-making are continuously recruited when juveniles live in social hierarchies, does the cumulative cognitive load of these interactions influence neurogenesis? The present study is the first to our knowledge to quantify neurogenesis in the telencephalon of juvenile *A. burtoni*. This is a particularly exciting question to ask, as fish have the most widespread adult neurogenesis of any vertebrate studied to date (Ganz and Brand, 2016). Neurogenesis in the telencephalon (forebrain) has been shown to be reduced in subordinates relative to dominants in fish (Sørensen et al., 2007), rodents (Gould et al., 1997), and primates (Gould et al., 1998). Prior research has linked reduced neurogenesis to exposure to chronic social stress (Tapia-De-Jesús et al., 2026). In juveniles specifically, social isolation reduces neurogenesis in marmosets (Cinini et al., 2014). Previous literature has quantified neurogenesis in adult *A. burtoni* (Maruska et al., 2012; Mack et al., 2021) and indeed observed that dominant and subordinate individuals differ in neurogenesis patterns across multiple regions (Maruska et al., 2012). In the present study we immunohistochemically labeled telencephalic tissue for doublecortin (DCX), a microtubule-associated protein that is used as a marker of differentiating neurons (Sánchez-Farías and Candal, 2015). While we were unsure if only a week of cohabitation would be sufficient to induce observable differences in neurogenesis patterns, we used this experiment as an opportunity to describe which brain regions exhibit neurogenesis in the juvenile *A. burtoni* brain and initially explore if these patterns reflected or differed from those reported in other studies on adults.

In summary, the present study assesses neural plasticity in a species that exhibits flexible social dominance interactions, adding to the rich literature on the formation and maintenance of juveniles' social hierarchies.

Materials and methods

Study organism

Sixteen *A. burtoni* juveniles were used from a laboratory population (see Hofmann/Renn lab strain (Pauquet et al., 2018)) approximately 65–70 generations descended from a wild-caught

colony (Fernald and Hirata, 1977). Adults were housed in mixed-sex breeding communities. Females lay eggs in a dominant male's territory, after which she scoops the eggs into her mouth, where the male fertilizes them. The mother orally incubates the developing larvae for 10–14 days. Experimental subjects were part of a larger cohort where we manually stripped the juveniles 12–14 days post-fertilization and placed them in community tanks (37.9 L) with other juveniles of similar age and developmental stage. The tanks contained a sponge filter, gravel substrate, and pieces of terracotta pots for habitat and territory. The fish used in this study were ~14 months old at the time of the experiment. To the best of our knowledge, these individuals were reproductively immature. Although fish of this age can be reproductively mature, development can be environmentally influenced (Fraley and Fernald, 1982), and in our pairs, no fish were included with nuptial coloration or anal fin spots, indicating male development (Fraley and Fernald, 1982), or signs of gravidity or oral incubation of eggs, indicating female development (White and Fernald, 1993). It is not feasible to sex juveniles until maturation; however, *A. burtoni* hatch at approximately a 1:1 female-to-male ratio (Heule et al., 2014).

The fish were fed once daily with Hikari Middle Larval Stage Plankton (Pentair Aquatic Eco-systems). All aquaria within the laboratory were maintained at an ambient temperature of 23.3–24.4 °C on a 12:12 light/dark cycle. All research was done in compliance with the Pitzer and Scripps Colleges Institutional Animal Care and Use Committee (IACUC Protocol # 22-006).

Experimental social groups

To form eight social pairs, fish were gently removed from the community tank using a hand net and placed into individual plastic beakers (200 mL) of aquarium water (Figure 1A). We paired larger (0.26 ± 0.033 g) and smaller (0.12 ± 0.0077 g) individuals randomly by size so that one fish was noticeably larger than the other to facilitate the rapid establishment of social status. For *A. burtoni* juveniles (Solomon-Lane et al., 2022) and adults (Alward et al., 2021), a larger body size benefits dominance. In this study, the larger fish were an average of 0.14 ± 0.03 g larger than their partners. Each pair was housed in a clear acrylic aquarium (26.7 long cm x 16.5 cm wide x 17.8 cm tall) with an air stone and no substrate. A small terracotta pot (6.35 cm diameter), broken to have an entrance, was placed at one end of the aquarium as a territory/shelter. White dividers prevented visual contact between different aquaria. The pairs remained undisturbed for seven days.

Social separation and reunion assay

On the eighth day, the pairs experienced a social separation, followed by a reunion. For the separation, each fish was gently removed from their experimental tank and placed individually into a clean glass beaker (20 mL) of aquarium water. White plastic posterboard was used to shield the sides of the beaker, as well as loosely cover the top, to prevent the fish from seeing outside visual disturbances. Fish remained separated, in their respective beakers, for 148–173 min (average 160.4 ± 3.1 min). To reunite the pairs, the larger and smaller fish were gently poured simultaneously from their beakers

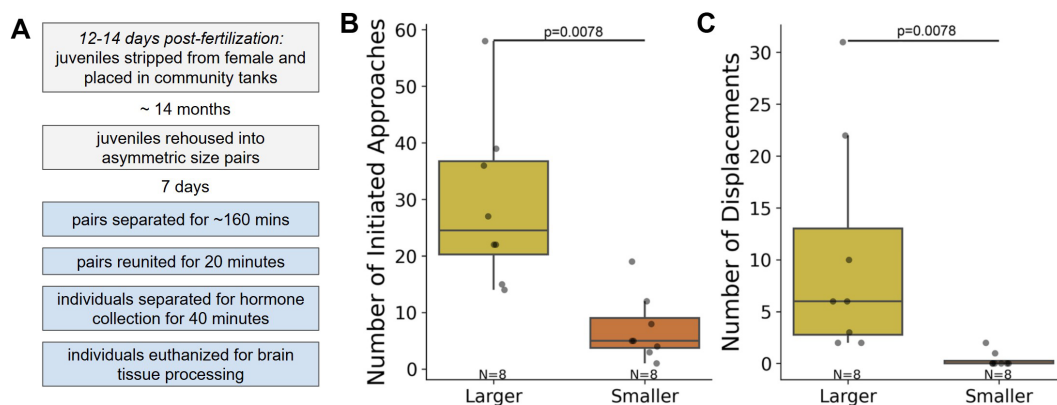


FIGURE 1

Fish were housed in pairs of asymmetric size for seven days and on the final day of the experiment (shown in blue boxes) were assessed for behavior, physiology, and neural responses following a separation-reunion assay (A). Larger fish ($n = 8$) exhibited significantly more territorial behavior than smaller fish ($n = 8$), evidenced by more initiated approaches (B) and displacements (C).

into their original experimental aquarium. Behavior was recorded from above (Warrior 4.0, Security Camera Warehouse, Asheville, NC), starting just prior to the reunion. The pairs remained together for 20 min. We then quickly removed the fish from their tanks using a hand net and placed them individually in glass beakers for water-borne hormone collection.

Videos of the behavior trials were scored (minutes 1–11 of the 20 min reunion) using Behavioral Observation Research Interactive Software (BORIS) (Friard and Gamba, 2016). We did not score the first minute as this is often when fish are simply reacting to being moved into the experimental tank and the experimenter's presence before they move out of the immediate area around the tank. We scored social approaches when one fish swam directly toward any part of the other fish's body, within three of its body lengths. A displacement was scored if the approached fish moved away from the approacher in any direction. Being displaced was counted as a submission. We also scored lateral displays, an escalated aggressive display where fish align head to tail and circle one another. For the territory, we scored how many times each fish entered the terracotta pot (with more than 50% of its body), and how long it remained. A fish exited the territory when its whole body was outside the terracotta pot.

Water-borne hormone collection

Fish were placed in clean 20 mL glass beakers that had been washed with soap and water, rinsed with DI water, rinsed with ethanol, and then rinsed with clean aquarium water. The beakers were then filled with 15 mL of clean aquarium water to collect water-borne hormones. The fish remained in the beakers for 40 min, shielded from visual disturbance with white plastic posterboard. It has been previously demonstrated in *A. burtoni* that during this time frame, steroid hormones from plasma are excreted from the gills, as well as urine and feces, into the water (Kidd et al., 2010; Friesen et al., 2012).

Brain collection

After the 40 min water-borne hormone collection, fish were gently removed from their beakers using a clean hand net for

euthanasia. Prior literature demonstrates that pS6 expression peaks anywhere from 60–150 minutes following exposure depending on the stimulus and brain region (Knight et al., 2012; Corrales Parada et al., 2024; Everett et al., 2025). Most relevant to our study, in Scaia et al. (2022), brain tissue from zebrafish exposed to social conflict were collected 45 minutes after the exposure, and the researchers observed significant differences in pS6 expression in regions of the SDMN. Notably, Scaia et al. conducted an experiment varying the post-exposure time course and did not see any differences when the wait time between social exposure and euthanasia was 30, 45, or 60 minutes (Scaia et al., 2022). Other studies on social behavior in fish have used even shorter wait times (30 minutes) and observed significant effects (Butler et al., 2018; Jackson et al., 2025). During the euthanasia procedure in our study, the fish were placed onto a paper towel wet with aquarium water and a digital photograph was taken next to a ruler to calculate body length (mm). Then, the fish were weighed to determine body mass (g). Immediately after body size measurements were collected, the fish were rapidly decapitated, and the whole head was placed into 4% paraformaldehyde at 4 °C for approximately 5.5 hrs (311–336 min, average 325 min). The whole head was then rinsed in a cold sterile 30% sucrose solution for 30 s, and then the brain was dissected out and placed into fresh 30% sucrose solution overnight at 4 °C. Afterwards, the brains were flash frozen in optimal cutting temperature (OCT) compound using dry ice and stored at -80 °C. Brains were shipped on dry ice to Amherst College for processing. At Amherst College, brains were cryosectioned into three identical series at 20 μm at -23 °C onto adhesive glass slides (Trubond 380 #50-340-33). Slides were stored at -80 °C until staining as described in Crescencio et al. (2024).

Immunohistochemistry and microscopy

Frozen slides were thawed for 5 min at room temperature, after which a hydrophobic barrier was drawn around the edge using an ImmEdge Hydrophobic Barrier Pen (NC9545623). Tissue was rinsed five times in 0.1 M phosphate-buffered saline (PBS) for 5 min and then blocked in with 0.3% Triton-X-100, 0.1% Tween, and 10% Normal Donkey Serum (NDS) in 0.1M PBS for 1 hr. Slices

were incubated overnight at 4°C in anti-phosphorylated ribosomal protein S6 (Mouse Monoclonal pS6, 1:500 Cell Signaling #62016, previously used in *A. burtoni* (Butler et al., 2018)) and anti-doublecortin (Rabbit Polyclonal DCX, 1:250, Thermo Fisher #481200) diluted with 0.3% Triton, 0.1% Tween, and 5% NDS in 0.1 M PBS. Sections were then rinsed twice in 0.1 M PBS for 30 min, and incubated in the dark in secondary antibody, (anti-mouse 488, 1:1000; Invitrogen #A21202, anti-rabbit 594, 1:1000 Invitrogen #A21207) diluted with 0.3% Triton, 5% NDS in 0.1M PBS for 2 hrs at room temperature. Following the secondary incubation, sections were rinsed twice in 0.1 M PBS for 20 min and then cover slipped using ProLong Gold Antifade Mountant with DNA Stain DAPI (P36931). A separate series of tissue from two fish in the present experiment was used as immunohistochemical controls with the identical procedure above but with either the primary antibody or secondary antibody omitted. Another separate series of tissue from one fish in the present experiment was double labeled for HuC/HuD (Mouse Monoclonal 1:200, Invitrogen 16A11 #A21271 followed by 1:1000 donkey anti-mouse 488 secondary) and DCX using the protocol described previously. HuC/HuD is a neuronal marker that has previously been labeled in *A. burtoni* tissue (Maruska et al., 2012).

Tissue was imaged using a Zeiss LSM980-Airyscan2 set to 20× magnification. To select appropriate sections, we referred to a telencephalic atlas published by Lauren O’Connell, Miles Fontenot, and Hans Hofmann (O’Connell et al., 2011). At least one hemisphere needed to be completely visible in the image and between 1–3 consecutive slices were imaged. Due to variability in tissue availability after extraction from the skull as well as tissue degradation during sectioning, 13 of 16 individuals (6 larger fish, 7 smaller) were successfully imaged and quantified. We selected brain regions of interest based on an initial qualitative assessment of interior (not ventricular) DCX expression and determined three brain regions of interest (ROI): the medial part of the dorsal telencephalon subdivision 3 (Dm3), the medial part of the supracommissural nucleus of the ventral telencephalon (Vsm), and the central part of the dorsal telencephalon (DC). We focused our efforts on interior, rather than ventricular, neurogenesis for two reasons: our tissue quality was less consistent in the ventricular areas and our preliminary qualitative assessment of interior DCX labeling identified neurogenic niches that were not yet well-described in prior literature for this species (Maruska et al., 2012).

The Dm3 is the putative homolog of the mammalian basolateral amygdala (Maruska and Fernald, 2018). The Vsm is the putative homolog of the extended amygdala, which encompasses the medial amygdala and the bed nucleus of the stria terminalis (Maruska and Fernald, 2018). The Dc does not demonstrate distinct homology to regions involved in the social decision-making network. It has previously been argued that the Dc should be considered as deeper layers of the respective nearby pallial regions, but this argument has limitations, as tracer injections to the Dc do not always align with the results of the nearby regions (Northcutt, 2011).

Images were uploaded into ImageJ and each channel was quantified independently. The brain region location was judged based on the previously described telencephalon atlas (O’Connell

et al., 2011). An oval ROI (ROI area (μm^2): DC: 17077.40, Dm3:44167.17, Vm3:16904.03) was placed on each individual region based on the morphological landmarks using the DAPI channel. Each region’s ROI was saved as an object and surface area was maintained across brain sections. These standardized ROIs were sized to only capture a central “punch” of the brain region of interest and not capture any tissue in adjacent regions. PS6+ labeled cells within each region were counted twice by one trained scorer (EA), blind to fish size category, using the ImageJ “CellCounter” plugin. The number of labelled cells reported is an average first across both counts of each section, then across sections, and lastly across hemispheres, resulting in one count per region per fish. Additionally, only cells expressing either DCX or pS6 where more than 50% of the cell body was located within the ROI were included in the cell count.

Solid-phase hormone extraction

We used 3 cc Sep-Pak Vac C18 columns (Water Associates, Milford, MA) on a vacuum manifold to extract the hormones and remove particulate matter. The columns were primed using two consecutive washes of 2 mL of methanol, followed by two consecutive washes of 2 mL of ultrapure water. The samples were run through the columns using vacuum pressure. Potentially interfering molecules were then washed using two consecutive washes of 2 mL of ultrapure water. We then dried the columns fully for 5 min using the vacuum. Hormones were eluted from the columns into 13 x 100 mm borosilicate test tubes with two rounds of 2 mL of methanol. The test tubes were then immediately transferred to a speed vac, where the samples were dried at 35 °C. Once dry, the samples were parafilmed and stored at -20 °C. The dry samples were first resuspended to 200 μL (5% ethanol, 95% ELISA buffer) and shaken on a multitube vortex for 30 min. The samples were then diluted 1:3 (50 μL sample into 150 μL ELISA buffer) and shaken again on the multitube vortex for 30 min. Water-borne cortisol was measured using the Cortisol Express enzyme-linked immunosorbent assay (ELISA) kit (Cayman Chemical, Ann Arbor, Michigan). All samples were assayed in duplicate, and the assay was carried out according to the kit instructions. The plate was read after 60 min, 75 min, and 90 min of development, and we present the results from the 90 min read. The standard curve was $r^2 = 0.99$. We assessed the coefficient of variation between the duplicates and found all samples were below 20% CV, thus all samples were included in the statistical analyses.

Statistical analysis

All statistical analyses performed can be referenced in [Supplementary Table 1](#), which reports the independent and dependent variables, p-value obtained, and statistical test used. All statistical analyses except the principal component analysis and calculation of the David’s score (described below) were performed in Python version 3.12.11. To assess relationships between categorical predictors and continuous response variables, a two-tailed two-sample T-test, Wilcoxon signed-rank test, or Mann-Whitney U test was performed

using the *scipy* module in Python (Virtanen et al., 2020). Which test used was determined by running a shapiro test for normality and whether the data was paired or unpaired. To assess how cortisol levels correlated to other metrics and how the difference in mass between the two individuals in the pair correlated to other metric we conducted, generalized linear models with no family specified using the *glm* function from the *statsmodels.formula.api* module in Python (Seabold and Perktold, 2010). For other continuous variables that fit a Poisson distribution (e.g. counts where the lower bound is zero) we performed generalized linear models with family set to Poisson using the *glm* function from the *statsmodels.formula.api* module in Python. A false discovery rate correction for multiple comparisons in pS6, and DCX expression within the 3 regions was conducted using the Benjamini-Hochberg method via the *multitests* function in the *statsmodels.sandbox.stats.multicomp* module in Python at $n=3$ for each marker. Specific statistical tests performed are listed in Supplementary Table 1.

Using the *pvcust* package in R version 4.5.1 conducted a principal component analysis (R Core Team, 2026; Suzuki and Shimodaira, 2006) to ask if an unbiased multidimensional analysis would significantly capture differences in size category. We included the four behavioral metrics (approaches, displacements, duration in territory, and lateral displays), average neurogenesis (doublecortin expression) across all regions, average neural response (pS6 expression) across all regions, and cortisol levels in the principal component analysis. We did not include doublecortin/pS6 expression for each region separately in the principal component analysis, as this removed a large number of individuals from the dataset for which we were unable to quantify expression in one of the three regions.

Using the *compete* package in R version 4.5.2 (Curley, 2016), we calculated an adjusted version of David's Score, a dominance index (Gammell et al., 2003), which we have previously used with juvenile *A. burtoni* (Solomon-Lane et al., 2022; Harmon et al., 2024). The adjustment incorporates both approaches and displacements, instead of categorizing all social interactions as either a "win" or "loss." Briefly, the agonistic outcome of individual *i* in interaction with another individual *j* is the number of times that *i* displaces *j*, divided by the total number of interactions (approaches) between *i* and *j* (i.e., *i* displaces *j* / (*i* approaches *j* + *j* approaches *i*)), allowing for the calculation of a dominance score in scenarios where dyads engage in equal numbers of repeated interactions. The rest of the calculations were completed using the *compete* package in R as described in Gammell et al. (2003) (Gammell et al., 2003).

Results

Size category significantly predicted behavior

Larger and smaller fish dramatically differed in their approaches ($p = 0.008$, $W = 0$, Cohen's $D = 1.97$; Figure 1B) and displacements ($p = 0.008$, $W = 0$, Cohen's $D = 1.31$; Figure 1C), but not lateral displays, time spent in the territory, or number of entries into the

territory. All larger fish had an adjusted David's score greater than 0, whereas all smaller fish had an adjusted David's score less than 0.

More asymmetric sizes produced more asymmetric behaviors

The relative difference in mass between the two individuals in the rearing pair significantly predicted the difference in approaches ($p < 0.0001$, $F = 32.7$, $R^2 = 0.98$; Figure 2A) and displacements ($p = 0.0015$, $F = 10.1$, $R^2 = 0.72$; Figure 2B) performed by the two individuals, but did not relate to the difference in lateral displays or time in territory.

Cortisol significantly correlated with behavior

Cortisol did not differ between larger and smaller fish. At an individual level, cortisol correlated to the amount of approaches ($p < 0.0001$, $F = 636$, $R^2 = 1$; Figure 3A) and displacements ($p < 0.0001$, $F = 491$, $R^2 = 1$; Figure 3B) but not to time spent in the territory, whether the fish entered the territory, or lateral displays. When categorizing the sixteen individuals into either "high" ($n = 8$) or "low" ($n = 8$) cortisol levels, approaches ($p = 0.047$, $W = 2$, Cohen's $D = -1.21$; Figure 3C) significantly differed between these two categories, while displacements ($p = 0.063$, $W = 1$, Cohen's $D = -1.00$; Figure 3D) showed a similar nonsignificant trend.

Behavior, but not size category, predicted overall neural response and neurogenesis

We observed areas of the brain where DCX overlapped with HuC/D and labeled cell bodies, confirming that DCX does indeed label neural-fated cells (Figure 4A). While other areas showed fibrous DCX expression, this expression did not have a morphology that would indicate labeling of radial astroglia. We observed widespread pS6 labeling in the telencephalon (Figure 4B). We quantified neural response in a subset of regions that also demonstrated interior (not ventricular) DCX expression (Figure 4C) to evaluate if regions of the brain active during social encounters were also particularly active sites of neurogenesis during development. Based on the observed expression patterns, we chose the Vsm, Dm3, and Dc for further analysis (Figure 4C).

No individual brain region demonstrated significant differences in neural response (pS6) or neurogenesis (DCX) between large and small fish following multiple test correction. Thus, we averaged all neural responses and neurogenesis data across the three regions quantified. With regards to behavior, global neural response (pS6) significantly correlated to approaches ($p < 0.0001$, $F = 61.20$, $R^2 = 0.99$; Figure 5A), displacements ($p < 0.0001$, $F = 32.09$, $R^2 = 0.93$; Figure 5C), and time spent in the territory ($p = 0.006$, $F = 7.57$, $R^2 = 0.38$), but did not correlate to lateral displays. Global neurogenesis (DCX) did not differ between large and small fish. But, like neural response, neurogenesis significantly correlated to multiple behaviors including approaches ($p < 0.0001$, $F = 17.09$, $R^2 = 0.73$; Figure 5B), displacements ($p < 0.0001$, $F = 26.66$, $R^2 = 0.95$;

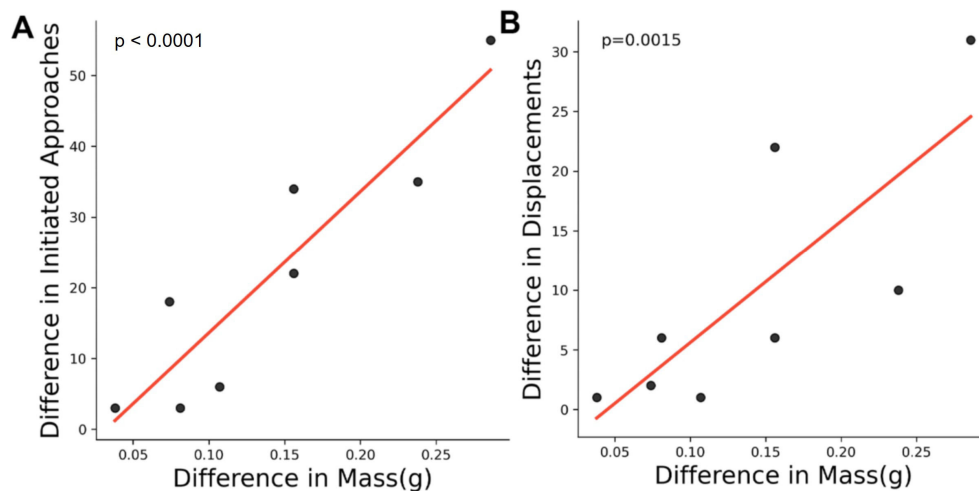


FIGURE 2

Fish were reared in pairs ($n = 8$ pairs, 16 individuals) of asymmetric size, but the difference in size between the larger and smaller individual varied across pairs. A greater size gap between the two individuals led to greater differences in initiated approaches (A) and displacements (B).

Figure 5D), time in territory ($p < 0.0001$, $F = 31.2$, $R^2 = 0.85$), and lateral displays ($p < 0.0001$, $F = 29.43$, $R^2 = 0.82$). Furthermore, both global neural response ($p < 0.0001$, $F = 18.37$, $R^2 = 0.70$; Figure 5E) and neurogenesis ($p = 0.0002$, $F = 13.65$, $R^2 = 0.59$; Figure 5F) exhibited a positive relationship with cortisol levels.

Multidimensional analyses differentiate individuals by size category

Principal Component 1 (PC1) explained 40% of the variance in the data and loaded most strongly by approaches (+0.486) and displacements (+0.497), whereas Principal Component 2 (PC2) explained 24% of the variance in the data and loaded most strongly by lateral displays (-0.650) and duration in territory (-0.506) (Figures 6A, B). Both PC1 and PC2 scores significantly differed between large and small individuals (PC1: $p = 0.041$, $t = 2.40$, Cohen's $D = 1.17$, PC2: $p = 0.038$, $t = -2.42$, Cohen's $D = -1.18$) (Figures 6C–E).

Discussion

As individuals undergo reproductive maturity, social relationships change and new behaviors emerge such as novel affiliative preferences (Baran et al., 2016), mating displays (Evans et al., 2002), and aggressive territorial defense (Calsbeek and Sinervo, 2002). In the present study, we assessed whether cohabitation in asymmetric dominance pairings as juveniles influenced the behavioral, physiological, and neural mechanisms of agonistic behavior in the cichlid *Astatotilapia burtoni*. Cichlids exhibit highly plastic dominance phenotypes in adulthood, thus the present experiment explored that plasticity by taking juveniles who had previously been housed in large breeding communities and pairing them in a size mismatched relationship. In doing so we found that this temporary

housing promoted divergent phenotypes in the larger and smaller individual.

We observed strong behavioral differences between dominant and subordinate juveniles. Larger juveniles were always the dominant individual in the pair, validating our and prior use of size mismatch to induce dominance relationships. Furthermore, a number of results supported the conclusion that each pair offers a distinct context that individuals must respond to. This pair-specificity is supported by our observation that the difference in agonistic behaviors between the larger and smaller individual directly correlated to their size difference. Prior research has also suggested that some pairs may be more “despotic” versus more “egalitarian” (Vehrencamp, 1983; Friesen et al., 2022), and when juveniles are size-matched with others they indeed show less hierarchical behavior (Solomon-Lane et al., 2022).

These dramatic differences in behavior between the larger and smaller individuals did not translate to differences in cortisol. In adult *A. burtoni*, Friesen et al. (2022) found that cortisol did not differ between dominant and subordinate adults, however Wallace et al. (2022) found that cortisol was higher in subordinates, and Maruska et al. (2022) found divergent conclusions across a review of multiple studies. Of note, Wallace et al. (2022) specifically measured cortisol in dominant males who had very recently attained territorial status in a group of eight individuals, suggesting there may be a temporary increase in cortisol arising from the specific increase in aggressive behaviors performed during the acquisition of territory, rather than a territorial vs nonterritorial state. Interestingly, paired-reared juveniles have been shown to exhibit different growth patterns than those reared in large groups (Solomon-Lane and Hofmann, 2019; Harmon et al., 2024), suggesting there may indeed be metabolic tradeoffs from these early life social environments that have not yet been fully detailed. However, based on the present results, this phenomenon seems unrelated to broad stress effects.

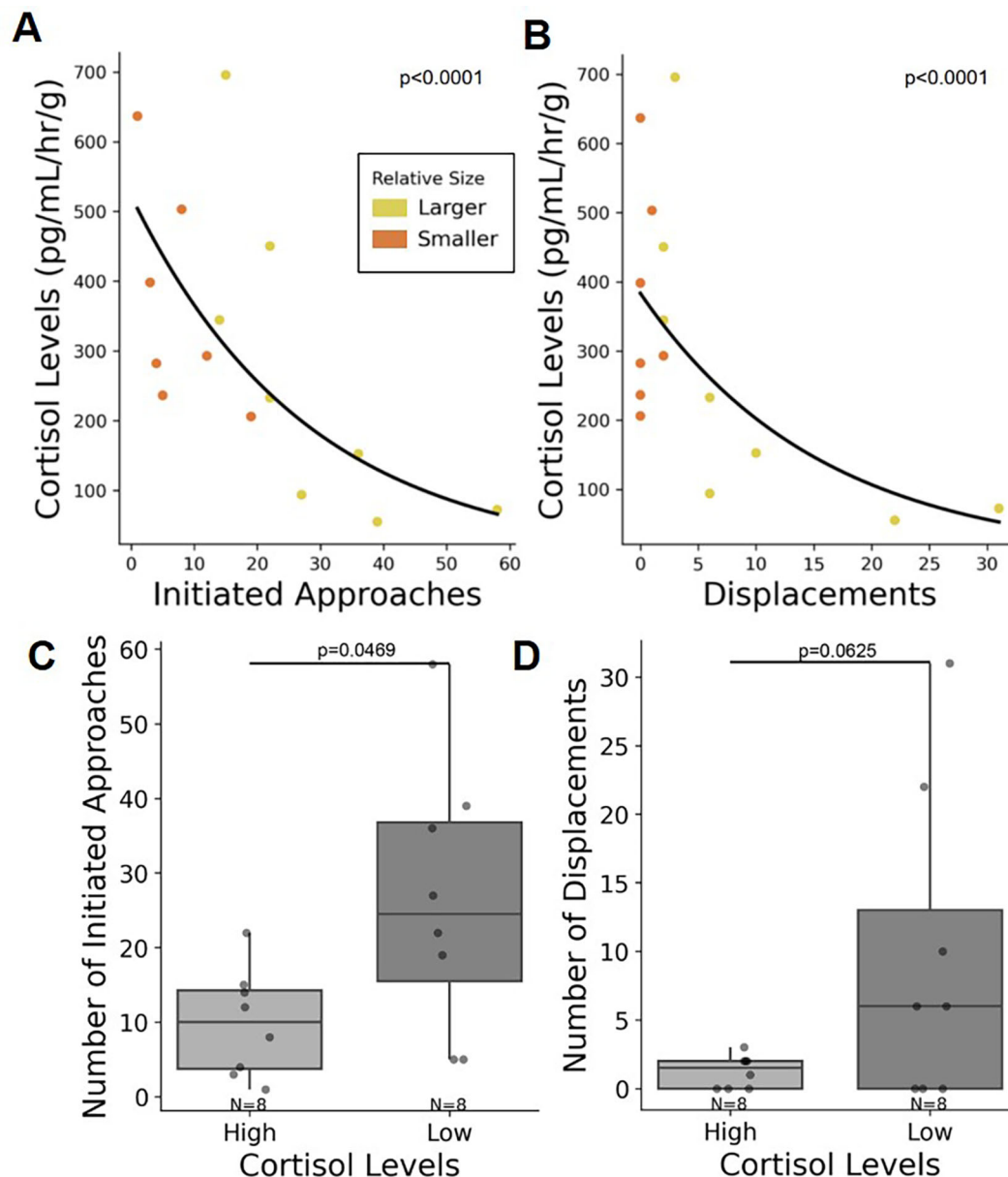


FIGURE 3

At an individual level ($n = 16$), cortisol levels were negatively correlated with the number of initiated approaches (A) and displacements (B) performed immediately prior to hormone collection. Individuals categorized in the “high” half of the distribution of cortisol levels ($n = 8$) exhibited fewer approaches (C) and displacements (D) than fish in the “low” half of the cortisol level distribution ($n = 8$).

We observed that, independent of size category, higher aggression is correlated with lower cortisol. This counters studies in other teleosts demonstrating elevated cortisol after territorial competitions (Ramos et al., 2021), but supports a prior finding in *A. burtoni* that fish who escalate aggression to attain a territory have lower circulating cortisol than those who maintain a consistent level of aggression (Alcazar et al., 2016). If individuals who are producing these aggressive behaviors demonstrate lower cortisol, we can reframe this interaction to suggest that individuals receiving these aggressive behaviors have higher cortisol. While receiving these aggressive behaviors might induce a temporary adaptive stress response, it does not seem to cause baseline state changes or cumulative effects in our study because (1) we did not see differences between large and small individuals, and (2) we saw cortisol correlate to approaches and

displacements but not to other behaviors measured. There is a rich literature supporting the conclusion that subordinates do not universally exhibit chronic stress responses (Creel, 2001; Abbott et al., 2003; Creel et al., 2013). Thus, we acknowledge that in our study subordinate status should not be simply described as a “negative” state that results from the accumulation of lost competitive interactions. Rather, we can reframe the subordinate state as one that either includes or is distinct from the initial “pre-territorial” state juveniles have when they first enter adult social communities.

The present study quantified neural response in the juvenile *A. burtoni* brain. We chose to quantify neural response in three regions (DC, Dm3, and Vsm) that showed the highest degree of interior neurogenic labeling, as we hypothesized that these may be sites where early life experience shapes neural trajectories that are

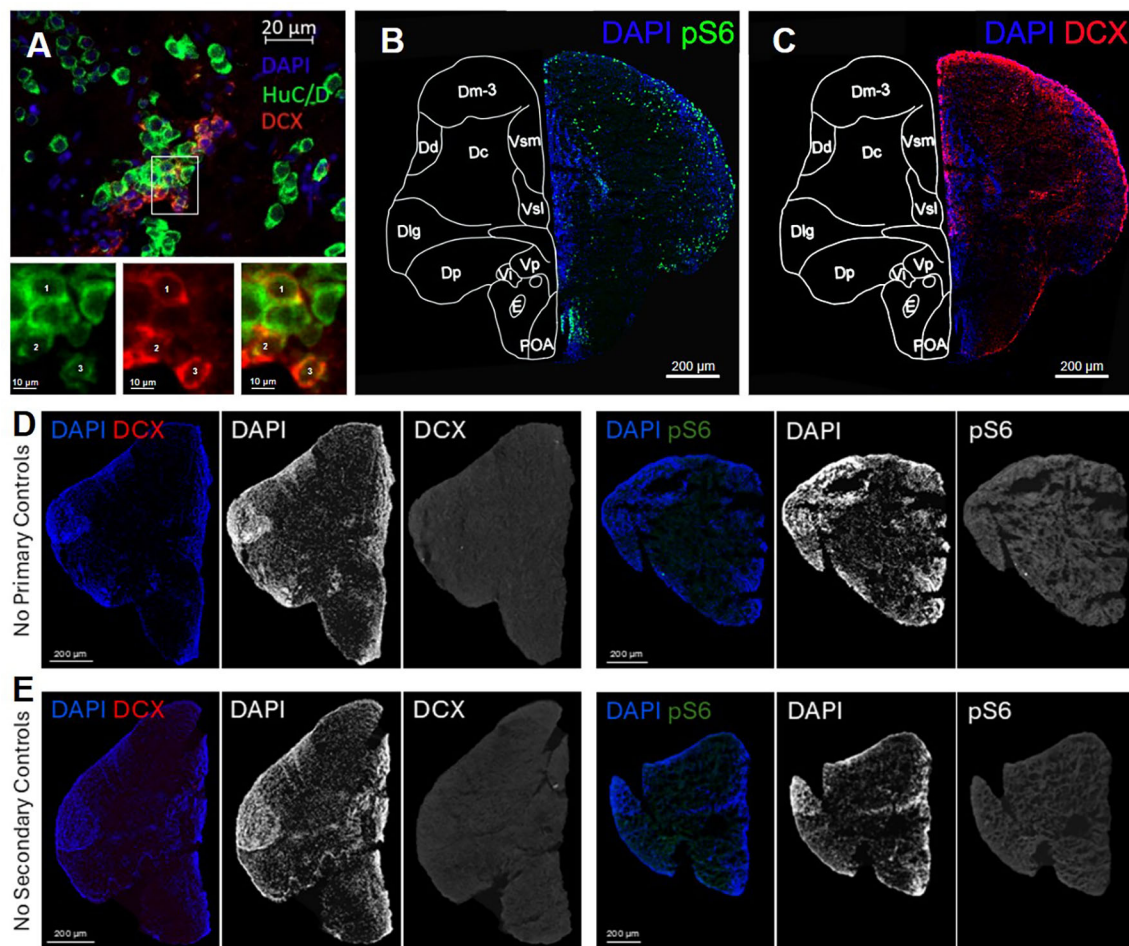


FIGURE 4

Doublecortin (DCX, red) labels newly-generated neurons in the fish telencephalon (A). DCX labeling demonstrated overlap with the neuronal marker HuC/D (green), confirming these DCX-labeled cells differentiate into neurons (A, bottom insert). Neural response was quantified using pS6, which showed widespread labeling (B). Similarly, while DCX showed widespread labeling (C), we focused on three sites that expressed neurogenic niches in tissue interior: the Vsm, Dm3, and Dc. A. *burtoni* neuroanatomy on the left hemispheres of panels B & C adapted from O'Connell et al., 2011 (O'Connell et al.). Control experiments that omitted the primary antibody (D) and secondary antibody (E) successfully inhibited labeling, with only background tissue autofluorescence visible even when the brightness and contrast of the specific channel was dramatically increased.

directly related to adult behavior. We found that a more global metric of neural response (averaging across our three regions) significantly correlated to individual approaches and displacements, likely reflecting the different social processing of the reunion context for each specific dyad. While we do not report the statistical results of individual brain regions given small sample size, we would like to highlight the Vs as a priority for future investigation. The Vs is homologous to the extended amygdala and is one of the few regions that shares bidirectional connections with both the Social Behavior Network and mesolimbic reward system (O'Connell and Hofmann, 2011). The Vs has been demonstrated to respond to different social contexts across multiple teleost species, including adult *A. burtoni* (Rodriguez-Santiago et al., 2022; Everett et al., 2025; Jackson et al., 2025). Of note, in adult New Zealand spotty wrasse (*Notolabrus celidotus*), Vs pS6 expression varies by dominance rank (Quertermous et al., 2025).

To assess our original overarching question of if the neural mechanisms underlying juvenile agonistic behavior is distinct from those employed by adults, we assessed qualitatively if the neural

responses observed in our juveniles reflected prior literature on adult *A. burtoni*. Friesen et al. (2022) found that subordinate males did not show increased neural response to an intruder in the Dm3 or the Vs (but did in other regions, including the teleosts' homologs of the lateral septum and hippocampus). Friesen et al. (2022) also found that dominant males showed increased neural response to an intruder in the Dm1 but not the other regions. This suggests that the combined neural responses during the reunion task observed in our three regions of interest (the Dm3, Dc, and Vsm) in juveniles may be distinct from the neural response of an adult to a novel intruder.

While we were unable to sex individuals in the present study, the sample in our experiment was presumably equally distributed between juvenile males and females. We believe that both male and female juveniles exhibited agonistic behavior and the associated neural responses in our study. In adulthood, females display male-typical social dominance behaviors and elevated androgens in the absence of males (Renn et al., 2012), emphasizing that all members of this species are capable of expressing agonistic social behaviors and forming status relationships. Of note, Jackson et al. (2025)

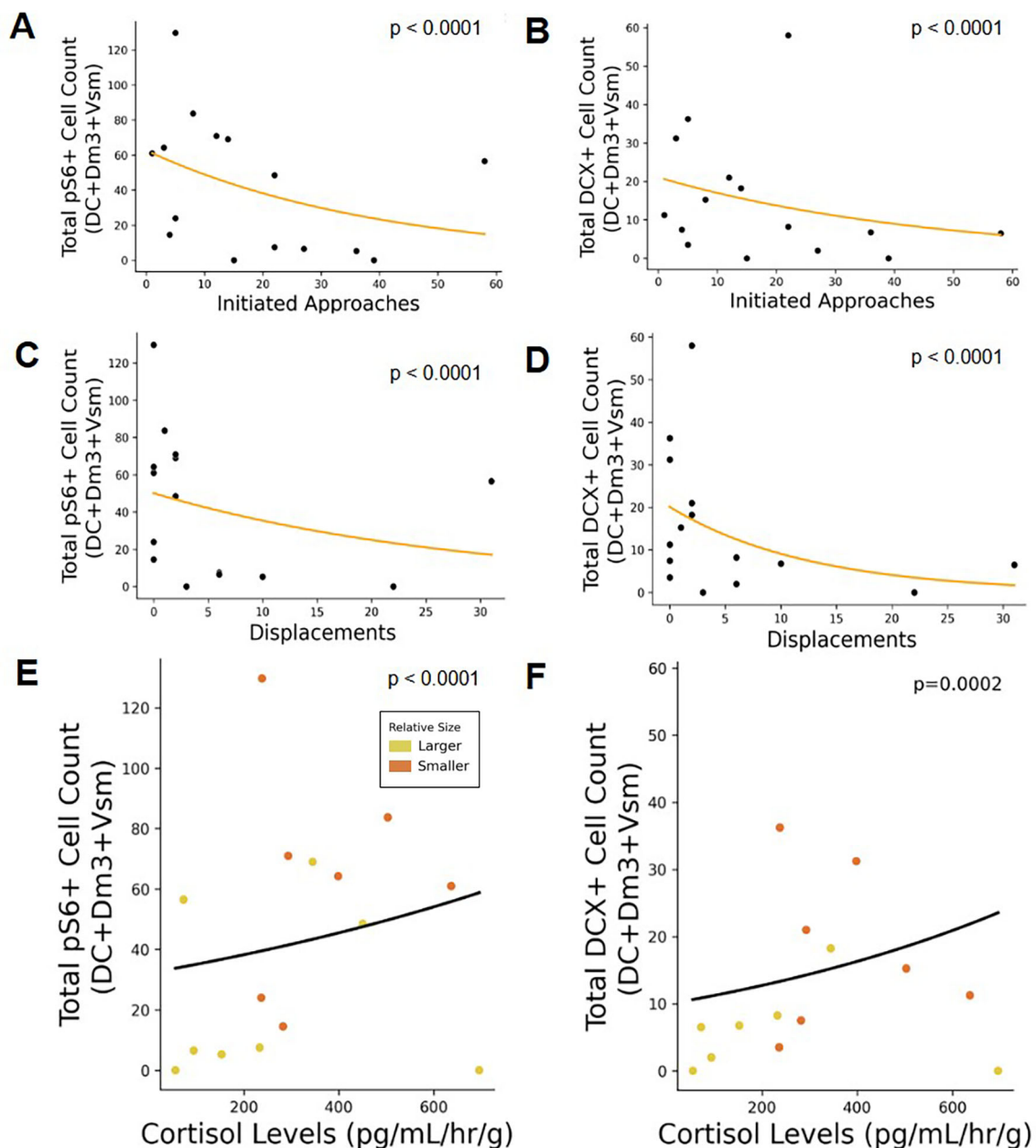


FIGURE 5

Neural response was quantified using pS6 in three regions of the telencephalon. When combining the pS6+ neuron counts of all three regions, global neural response significantly correlated to an individual's ($n = 15$) approaches (A), displacements (C), and cortisol levels (E). Similarly, neurogenesis quantified using DCX in three regions of the telencephalon correlated to an individual's ($n = 15$) approaches (B), displacements (D), and cortisol levels (F).

observed no behavioral sex differences when *A. burtoni* were presented with their mirror image and no sex differences in neural response in the Vs and Dm3 (Jackson et al., 2025 did not quantify the Dc as we did). Thus, we presume our results are not reflective of unaccounted-for sex differences, but sex would indeed be a promising avenue to explore in the future.

Here we report the first characterization of neurogenesis in the juvenile *A. burtoni* telencephalon. The morphology of the doublecortin-labeled cells and the colabeling of doublecortin and HuC/D supports that these are neural-fated cells. To confirm these are not radial astroglia-fated cells, future research should co-label for markers such as glial fibrillary acidic protein (GFAP) or glutamine synthetase (GS). In the adult *A. burtoni*, telencephalon

DCX and GS do not co-label (Mack et al., 2021), and an investigation of noninjury-induced neurogenesis in juvenile sharks found that DCX and GFAP do not colocalize, but the relevance to the present study is limited as this was in the retina (Sánchez-Farías and Candal, 2015).

The present study is the first to characterize neurogenic niches in the juvenile *A. burtoni* telencephalon. We identified the Dc as a particularly active site of interior neurogenesis, which corroborates an area highlighted as a source of DCX-labeled neurogenesis in adult *A. burtoni* (Mack et al., 2021). This observation that the Dc has different neurogenesis patterns than nearby regions (with the exception of the Dm3, which also showed an interior neurogenic population) lends support to

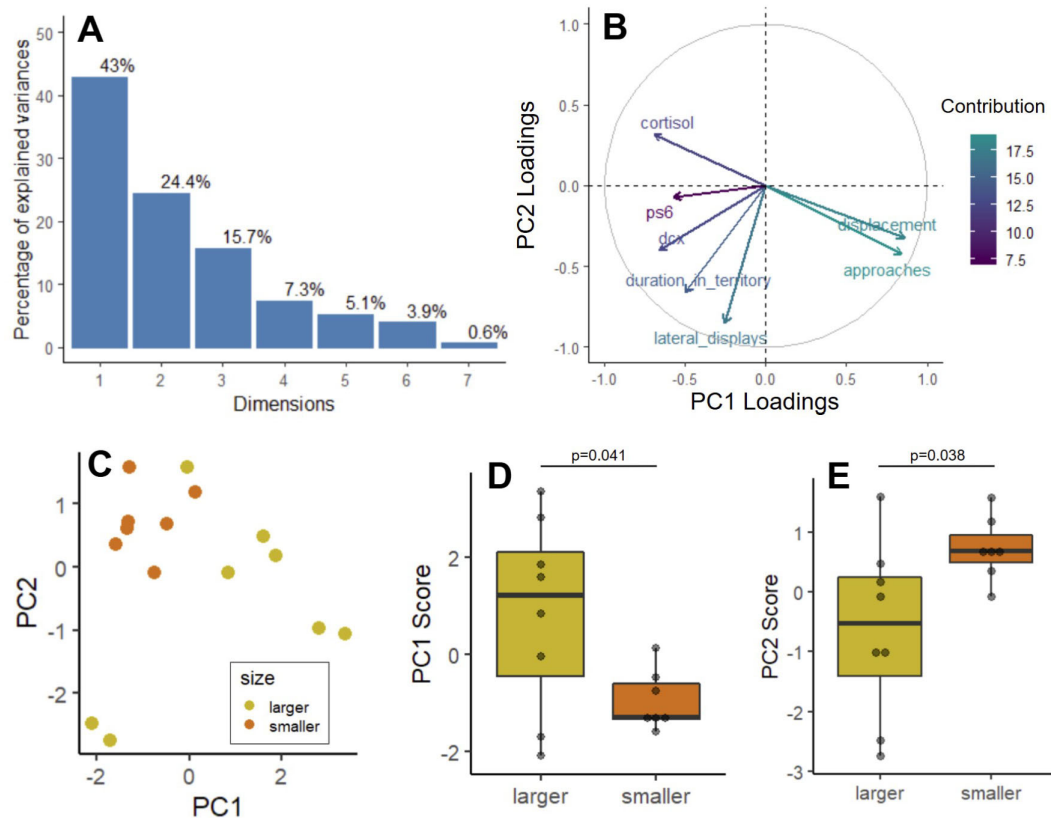


FIGURE 6

A principal component analysis combining behavioral, physiological, and neural measurements found that the first two components explained over 50% of the data; thus, PC1 and PC2 were selected for further analysis (A, B). Both PC1 and PC2 scores significantly differed between larger and small fish (C–E).

the hypothesis that the Dc should be considered a distinct region in the telencephalon and not simply extensions of nearby regions (Northcutt, 2011), though further work is needed. If we more indirectly compare juvenile DCX expression to adult BrDU expression (another marker of neurogenesis), we see that Maruska et al. (2012) also observed interior neurogenesis in the Dm3 and Dc. However, Maruska et al. (2012) did not highlight the Vsm as an area with interior neurogenesis; rather, they observed that BrDU labeling in the Vs was restricted to the medial ventricular border. This suggests that the Vs may be an area of interest for explicit studies comparing juvenile and adult neurogenesis. Lastly, while we focused our study on telencephalic neurogenesis, we did observe during our tissue processing that there is a large amount of DCX expression in the optic tectum (as reflected in Maruska et al., 2012).

In our study, neurogenesis was not influenced by short-term cohabitation but did negatively correlate to aggressive behavior at an individual level. It is not surprising that we further identified a relationship between neurogenesis and cortisol levels. In mice, chronic social defeat reduces hippocampal neurogenesis but interestingly this suppression of neurogenesis can be blocked via administration of a glucocorticoid receptor antagonist (Mouri et al., 2018). The unexpected result is the directionality of these relationships, where in our study fish with low approaches, displacements, and cortisol demonstrated the highest neurogenesis.

This result reinforces the hypothesis that instead of characterizing classically subordinate or “defeated” individuals, we may instead be capturing a specific “pre-territorial” state that juveniles progress through prior to acquiring their first territory.

Upon evaluating the individual behavioral, physiological, and neural variables in our study, we combined them into a principal component analysis to assess if an unbiased multidimensional analysis would distinguish our large and small fish. The principal components supported the conclusions from our other statistical analyses that behavioral metrics are the best predictors of size category, not neural or physiological variables. We qualitatively noted that larger individuals showed more variation in PC scores than small individuals, again supporting that the way to “be dominant” might vary from pair to pair based on either the size difference or other factors. This variation is supported by Friesen et al. (2022), who found that dominant males show a high degree of individual variation in their space use.

To conclude, here we observed that in juvenile cichlids cohabitation in a size-mismatched pairing induced a dominance relationship that can be observed via behavior and neural response. When comparing across pairs, the degree of size difference predicted asymmetry in behavior. When comparing across individuals regardless of pairing status, physiology, neural response, and neurogenesis correlated with behaviors expressed at an individual level. While we did not explicitly compare juveniles and adults in the present study, this work helps us better

understand the ontogeny of the mechanisms of social dominance in this species. Our behavioral results support the conclusion that the degree of behavioral plasticity exhibited in adults is also present in the juveniles. Our neural results suggest that the social processing driving juvenile behavior is distinct from the social processing of adults. Our pairing was only one week and our individuals had previously been housed in complex social groups, thus our results likely reflect plastic responses to social contexts rather than a particular time window permanently altering phenotypes. Taken together, this work suggests that each pairing produces its own unique social dynamics that these fish, even as juveniles, are highly attentive to.

Data availability statement

The datasets presented in this study can be found in online repositories. The names of the repository/repositories and accession number(s) can be found below: https://github.com/kellyjwallace/Aquino_2026_Front_Ethol.

Ethics statement

The animal study was approved by Institutional Animal Care and Use Committee of Pitzer and Scripps Colleges. The study was conducted in accordance with the local legislation and institutional requirements.

Author contributions

EA: Investigation, Visualization, Data curation, Writing – original draft, Formal analysis, Methodology. ES: Writing – review & editing, Investigation, Data curation. AR: Data curation, Writing – review & editing. JS: Methodology, Writing – original draft, Validation. TB: Writing – original draft, Validation. TS-L: Writing – review & editing, Conceptualization, Writing – original draft, Formal analysis, Methodology, Resources. KW: Methodology, Formal analysis, Writing – review & editing, Writing – original draft, Conceptualization, Resources.

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Supplementary material

The Supplementary Material for this article can be found online at: <https://www.frontiersin.org/articles/10.3389/fetho.2026.1888727/full#supplementary-material>

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